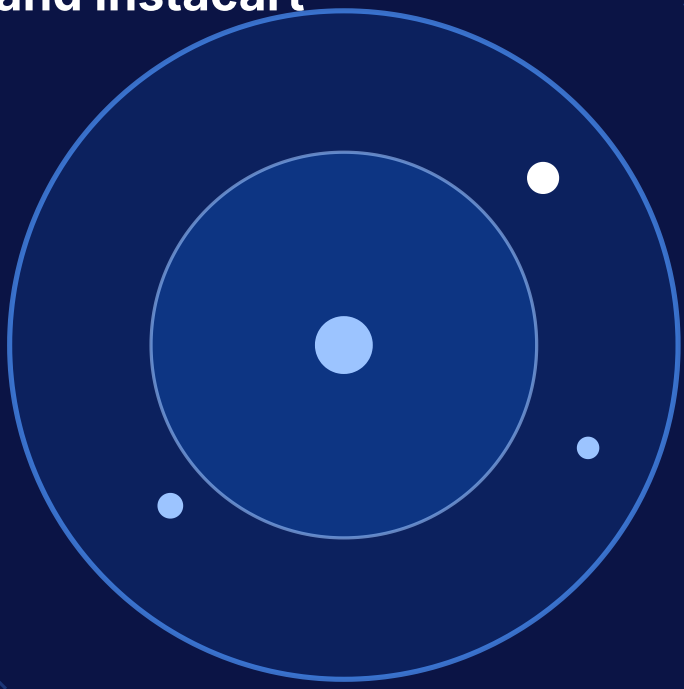
Agent-Ready Instacart

When a customer's own AI assistant shops for them, **the customer sets the rules and Instacart enforces them.**



Q1 Where to start

Q2 The board demo

Q3 Which API first

Q4 "Quantity: 2"

Q5 How much autonomy

Q6 Should we worry?

Varun Kale

ED22B018 · Indian Institute of Technology Madras

Instacart case study

Six product decisions, one live prototype

AGENDA

TAP ANY ROW TO JUMP

Six questions, one story.

An outside AI assistant is shopping on Instacart. In every answer, **the customer sets the rules and Instacart enforces them**: each action is Act, Ask or Block.

EXECUTIVE SUMMARY
00 **The whole argument on one page** — p3
The brief, the risk, the answer and the six calls

Q1 · WHY GROCERIES
01 **Start with the weekly basket** — p4
Frequent, judgment-heavy and cheap to fix

Q2 · THE BOARD DEMO
02 **Keep the date, change the demo** — p5
Show the rules the pilot will actually run on

Q3 · CHECKOUT OR SUBSTITUTIONS
03 **Substitution preferences first** — p6-7
Judgment before speed; the customer pays for now

Q4 · "QUANTITY: 2"
04 **Fix the contract, not the agents** — p8-9
An explicit unit and a preview before payment

Q5 · ORDERS PER USER PER MONTH
05 **4 per rolling 30 days, then test** — p10-11
A capped, randomized pilot creates the missing data

Q6 · "SHOULD WE BE WORRIED?"
06 **It could. Here is how we'd know early** — p12
Gates with owners, agreed before launch

PROTOTYPE AND PLAN
07 **See it work, then the 90 days** — p13-14
Live prototype, roadmap and scorecard

APPENDIX
08 **Sources and how each number is worked out** — p15-16
Every external figure, one tap from its source

Varun Kale · ED22B018 · IIT Madras

Live prototype: **inmobi-prototype.vercel.app**

EXECUTIVE SUMMARY · THE WHOLE DECK ON ONE PAGE

The customer sets the rules. Instacart enforces them.

Agents won't fail because they can't click "checkout". They fail by acting confidently outside what the customer meant. So we ship **trust first, autonomy second**.

01 THE BRIEF

Make Instacart usable by an assistant the customer brought

Within **12 months** agents will compare, order, cancel and adjust for users, often with nobody checking each step. The assistant shipped with the phone, not with Instacart.

02 THE REAL RISK

Confidently wrong, not slow

Already **1 in 5** online grocery shoppers get a substitution. An agent repeats a wrong swap or a misread "2" every week, and one viral "AI ordered the wrong groceries" story outweighs months of growth.

03 OUR ANSWER

Rules live at Instacart, every assistant reads them

Saved rules + an explicit contract + a decision for every action: **ACT** **ASK** **BLOCK**.
Autonomy is earned with data, never granted upfront.

04 SIX QUESTIONS, SIX CALLS

Each row is one page of evidence. Highlighted numbers are the ones to remember.

Q	THE QUESTION	OUR CALL	WHY, IN ONE LINE	WE'LL WATCH · INITIAL TARGET
Q1 p4	Why start with groceries?	**Start with weekly repeat orders.** Familiar baskets make every change visible.	**~52** shops a year, each mistake is one refundable line.	Review time vs regular reorder; repeat delegation
Q2 p5	Flashy board demo or push back?	**Push back, but don't delay.** Keep the date; demo a real stockout with customer review.	The shortcut breaks within **3 months** of real agent traffic.	All agreed demo cases pass; **≥70%** of demo work reused
Q3 p6-7	Checkout or substitution API?	**Substitution preferences.** The customer keeps paying in the Instacart app until Q5's test has data.	Judgment lives in stockouts; a customer tap at payment is the safety net we want early.	Swap acceptance **≥80%**
Q4 p8-9	Why did "quantity: 2" differ?	**Our contract lost the unit.** Require SKU + unit; preview before payment.	Same words: **$3.60** for 1 kg vs **$6.40** for 2 kg.	**0** silent mismatches in shared tests
Q5 p10-11	Orders per user per month?	**4 per rolling 30 days**, max 1 per approved weekly window, within the spend cap.	Matches a weekly habit; caps exposure at **$320**.	Unintended orders **≤0.5%**, at 95% confidence
Q6 p12	"Will this happen to us?"	**"It could. Here's how we'd know early."**	Owners, evidence and pause rights agreed before launch.	Every gate has an owner and a decision

05 THE STORY IN FIVE MOVES

Same order as the deck

Q1 · ARENA
Pick the weekly shop
Frequent, repeatable, cheap to fix

→ Q2-3 · PRIORITISE
Build the rules first
Demo and sprint on what lasts

→ Q4 · INTERPRET
Make the unit explicit
One contract for all agents

→ Q5 · AUTONOMY
Earn the right to order
Cap, test, then grow

→ Q6 · LEAD
Say what we'd watch
Gates with owners

WEEKS 1-4
Substitution rules

WEEKS 5-6
Unit contract + board demo

WEEKS 7-12
Customer-checkout pilot

THEN, IF GATES PASS
Capped-autonomy test

06 SEE IT WORK

Live prototype

Maya's Sunday basket, end to end

A real stockout (**Act $4.50 · Ask $5.50 · Block regular milk**), an ambiguous "2", and a customer-placed order of **$70.77**, **$9.23** under her cap.

inmobi-prototype.vercel.app · 60-s autoplay

60 s autoplay, hands-free

5 stops guided tour, you tap Ask

3 layers request, rule, verdict

07 SCOPE WE COMMIT TO IN THE FIRST 90 DAYS

- **One store, packaged staples**: no loose-weight produce until the unit contract is proven.
- **Assistant prepares, customer places**: autonomous ordering is a separate, opt-in test.
- **Rules are customer-owned**: agents can read them, not edit them, in the pilot.
- **Two agent platforms** integrate against one shared, versioned contract.

08 ASSUMPTIONS

- The assistant is a **third party** (shipped with the phone) connecting through Instacart's API or MCP.
- Prices, baskets and pilot dashboards are **illustrative**; targets are proposed pilot settings.
- There is **no history** of agent-placed orders (per the brief), so Q5 creates the data first.
- Q3 is taken as written: **neither API is agent-readable yet**, so we choose one.
- External sources are listed in the Appendix; anything unsourced is reasoning.

Q1 · MAKE THE CASE FOR GROCERIES

Weekly groceries let us test judgment on a basket the customer already knows.

THE CALL
Start with repeat customers and their usual weekly basket.

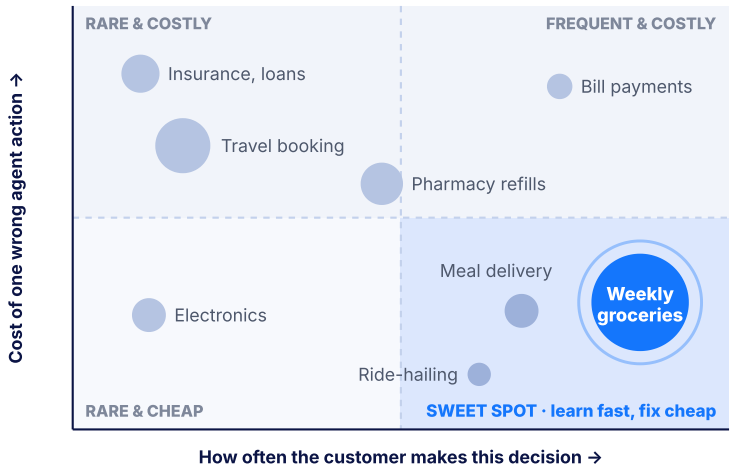
WHY
Frequent, repeatable decisions where each mistake is **1 line**, visible at the door and refundable.

TRADE-OFF ACCEPTED
Taste and freshness are subjective, so per-item rules come first (Q3).

WE'LL WATCH
Review time vs a regular reorder; share of customers who delegate again.

01 WHERE SHOULD AGENTS LEARN FIRST?

Directional, my judgment



Bubble size = judgment needed per order (swaps, sizes, multi-item trade-offs). Meal delivery is as frequent, but one restaurant is one decision.

WHY NOT TRAVEL
Rare, and one wrong booking is costly and hard to undo.

WHY NOT BILLS
Frequent, but little judgment: nothing to learn from.

WHY GROCERIES
Frequent, judgment-heavy and cheap to fix: refund one item, not the trip.

02 WHAT THE EVIDENCE SAYS

Sources in appendix

~52
shopping occasions a year for a weekly customer: **52 chances to learn**

1 in 5
UK online grocery shoppers got a substitution in their latest order (**29%** the year before)

INSTACART FIELD EXPERIMENT · 840,000+ CUSTOMERS

-2.9% product replacements
-7.5% poor replacement ratings
-3.7% refunds

...just from **telling customers stock was low**. Stockout handling measurably moves satisfaction on Instacart itself.

03 THE ANSWER IN ONE LINE

A familiar basket makes every change visible. A stockout tests whether the assistant follows a **saved rule**, cheaply and every week.

04 FIVE REASONS THE WEEKLY BASKET WORKS

Each reason, and what it means for the pilot

1 Cadence
A similar trip every week gives another chance to see what works.
→ COMPARE WEEK ON WEEK

2 Baseline
The saved list exposes a changed item, size or price instantly.
→ REVIEW ONLY CHANGES

3 Judgment
A missing staple shows which preferences the assistant can apply.
→ TEST REAL CHOICES

4 Contained
Packaged items and customer checkout keep the first test bounded.
→ LIMIT EXPOSURE

5 Habit
Saved rules carry forward, so each order needs less setup.
→ TRACK REPEAT USE

05 FIT BY JOB IN THE BRIEF

Illustrative fit

JOB IN THE BRIEF	REPEATS	JUDGMENT	UNDO
Rebuild the weekly basket	●	◐	●
Maintain the recurring list	●	○	●
Handle a substitution	◐	●	●
Confirm and place the order	●	◐	○
Cancel or adjust after placing	◐	◐	●

● High ◐ Medium ○ Low · **Highest judgment: substitutions → build them first (Q3)**

06 WHAT CAN GO WRONG, AND WHERE THE DECK HANDLES IT

RISK	HANDLED BY
Wrong replacement · diet, taste, brand	Q3 Per-item rules; block hard mismatches
Wrong quantity · each, pack or weight	Q4 Explicit unit + checkout preview
Unwanted order · whole basket at risk	Q5 Customer checkout first, then a cap
Support cost · thin grocery margins	Q5 Refunds + contacts per order

Q2 · THE FLASHY BOARD DEMO

Push back, but don't delay. Keep the board date; change what we demo.

THE CALL
Keep the 6-week date. Demo a real stockout with customer review instead.

WHY
The requested shortcut is likely thrown out within **3 months** of real agent traffic. The rules demo becomes the pilot.

TRADE-OFF ACCEPTED
A narrower, less flashy story for the board.

WE'LL WATCH
All agreed demo cases pass; **≥70%** of demo work reused in the pilot.

01 THE ASSUMED REQUEST

Brief leaves it open

"One-tap reorder: the assistant re-buys the whole basket, no review."

Looks great on stage. Under real agent traffic it fails three ways:

1 · Stockout with no saved rule
The assistant guesses a swap. Wrong item, at the door.

2 · Price moves after the quote
The basket silently goes over the customer's cap.

3 · Agent retries a timed-out call
No order ID, so a retry places a **duplicate order**.

02 WHAT THE BOARD SEES IN WEEK 6

Controlled demo · illustrative prices

"Get my usual Sunday basket for $80 or less. If my milk is out, lactose-free up to $5 is fine."

✓ ACT
$4.50 lactose-free
Inside the rule. Added, reason shown.

◇ ASK
$5.50 lactose-free
$0.50 over the limit. One-time OK.

× BLOCK
$3.29 regular milk
Breaks the lactose-free rule.

THEN
"Wheat ×2" is held: 2 × 1 kg = 2 kg, **$6.40** previewed →

THEN
Cap checked: **$70.77**, $9.23 under $80 →

FINALLY
The customer places it, live, on stage

Every step runs in the live prototype. **The demo is the pilot's first test case**, not a throwaway.

03 THREE OPTIONS FOR THE BOARD REQUEST

TEST	BUILD AS ASKED	PUSH BACK AND DELAY	✓ KEEP DATE, CHANGE SCOPE
Week 6	Shows the original feature	Misses the board checkpoint	**Shows both swaps, the unit check and customer checkout**
Month 3	Rework once live traffic exposes the gaps	Stronger build, feedback comes later	**Same rules and cases re-run on pilot traffic**
Sponsor	Happy now, blamed later	Loses momentum and trust	**Keeps the date and a credible story**
Verdict	× Throwaway work	× Lost checkpoint	✓ Recommend

04 SIX WEEKS TO THE BOARD MEETING

WEEK 1
Verify the handoff
Prepared cart reaches customer checkout with the right items and total.

WEEKS 2-4
Build and test the rule check
Saved milk rule, Act / Ask / Block verdicts with reasons, the unit hold for "2", both swap cases on two test adapters. Same sprint as Q3.

WEEK 5
Rehearse failure
Run a failed handoff; record the fallback.

WEEK 6
Show and ask
Run the demo; ask for a customer-checkout pilot.

05 WHAT I'D TELL THE SPONSOR

"We keep the date. Let's show the board when the assistant can choose a swap and when it has to ask. Those are the same rules the pilot will run on."

06 WHO DECIDES

Instacart product lead and our **engagement lead** sign off on scope, risks and the fallback in a one-page risk note.

Agreed in week 1, so it's not a fight in week 5.

07 IF I'M OVERRULED

- Build it as a **time-boxed staging mock-up**, not production.
- Record what we expect to break.
- Protect the 4-week rules sprint; book a **month-3 review**.

Q3 · 2 ENGINEERS, 4 WEEKS: CHECKOUT API OR SUBSTITUTION-PREFERENCE API?

Substitution preferences first. That is where trust is won or lost.

THE CALL
Make substitution preferences agent-readable. The customer pays in the Instacart app for now.

WHY
Judgment is the hard part. A customer tap at payment is the safety net we want while there is no agent-order data (Q5).

TRADE-OFF ACCEPTED
Agents can't finish the weekly order alone yet, so order lift waits a cycle. Slower is safe.

WE'LL WATCH
Swap acceptance **≥80%**; drop-off at the customer-pay step.

01 SAME 8 ENGINEER-WEEKS, TWO CHOICES

Neither is agent-ready today

CRITERION	SUBSTITUTION PREFS	CHECKOUT
Handles the hardest moment (a stockout)	●●●	●○○
Applies the customer's boundaries	●●●	●○○
Safe while the other stays human-only	●●●	○○○
Lets the agent finish the order alone	○○○	●●●
What the other gap leaves	Customer taps to pay: slower, reviewed	Stockouts guessed, then paid automatically
Smallest useful build	Read 1 saved rule, judge 1 candidate	Quote, fees, place order, errors
What we learn	Do customers keep proposed swaps?	Do agents complete orders?

Checkout wins only on finishing the order alone: the step we want a human on until Q5 has data.

COST OF BEING WRONG · A REGRET CHECK

IF WE PICK...	...AND THE TRUTH IS	WE LOSE
Substitutions	Customers drop off at the pay step	1 cycle of order lift; checkout is next sprint, built on the rules
Checkout	Stockout judgment breaks trust	Wrong orders placed automatically, then rules retrofitted into a live path

Smaller, reversible worst case: a delay, not a wrong order at the door.

02 WHAT EACH CHOICE LEAVES THE CUSTOMER

✓ BUILD SUBSTITUTIONS
Agent fills the cart and settles stockouts
The cart opens in the Instacart app; the customer reviews and pays. We call this the **handoff**.

BUILD CHECKOUT
Agent can pay, but can't judge
Every stockout is a guess or a stop, and the guess is already paid for.

Real-world check: Instacart already lets third-party AI apps hand a cart to customer checkout (p16), so this path is realistic.

STILL MISSING FOR AGENTS

- A machine-readable rule an outside assistant can read
- A verdict with a reason it can relay to the customer
- A rule version, so a stale rule is never applied

03 WHY JUDGMENT COMES FIRST

Stockouts are where the assistant must act like a **trusted personal shopper**, the brief's own words. In Instacart's own experiment, better stockout handling moved:

-2.9%	-7.5%	-3.7%
replacements	poor ratings	refunds

Paying without the customer's limits ships wrong orders faster: Q5's exact complaint.

04 STEELMAN: "ORDERING IS THE JOB, SO CHECKOUT FIRST"

The case for checkout: placing the weekly order is the brief's first job, and agent orders are the growth leadership counts.

Why I still defer it: paying without judgment turns each stockout into a paid mistake. A tap costs seconds; a wrong order costs trust. Checkout is sprint two.

Honestly, it is close. For order volume, checkout wins. The brief's risk is complaints.

05 WEEK-1 GATE: WHEN I'D FLIP

Evidence, not opinion

Send an agent-built test cart into the Instacart app. Check store, exact lines, fees and total; the customer places one order.

✓ CART ARRIVES INTACT
Build substitutions
The customer can pay; rules have an order to protect.

× NO WAY TO PAY
Switch to a minimal checkout
Rules need an order to act on: build a reviewable quote first.

INITIAL TARGET
80%+
swap acceptance

06 HOW WE MEASURE SUCCESS

Swap acceptance
Proposed swaps the customer keeps at checkout.

Edits per stockout
How often the customer fixes a swap or answers a prompt.

Pay-step drop-off
Agent-built carts the customer never pays for.

Review time
Seconds to approve vs a manual reorder.

Q3 · THE FOUR-WEEK BUILD

Two engineers can ship one rule check in four weeks.

THE SCOPE
One store, packaged staples, rules the customer has already saved.

WHY THIS SMALL
One verdict API, used by every agent, beats four half-built features.

DEFERRED
Agent checkout (sprint two), agents editing rules, swaps after checkout.

RELEASE GATE
Both adapters pass **100%** of scoped cases. Any rule-breaking swap blocks rollout.

01 THE PROPOSED API, IN ONE CALL

Illustrative

The assistant asks: **"Can I use this replacement?"** Instacart answers Act, Ask or Block, with a reason.

REQUEST · ASSISTANT → INSTACART

```
{"cart_id":"draft_17", "line_id":"milk_01",
 "candidate_sku":"lf_1l"}
```

CUSTOMER'S SAVED RULE · READ BY INSTACART

```
{"requires":"lactose_free", "pack_ml":1000,
 "ask_above_cents":500, "rule_v":3}
```

VERDICT ON THE $4.50 CANDIDATE

```
{"decision":"ACT", "reason":"WITHIN_RULE", "rule_v":3}
```

EDGE RESPONSES THE ASSISTANT MUST HANDLE

```
{"decision":"ASK",   "reason":"RULE_MISSING"}
{"decision":"BLOCK", "reason":"RULE_CONFLICT"}
{"decision":"ASK",   "reason":"STALE_RULE_VERSION"}
```

Instacart re-checks live price and stock for that SKU at cart update and at checkout.

02 HOW INSTACART DECIDES

Same rule, any agent

The usual milk is out. Each candidate is checked against the customer's saved rule:

✓ ACT — **Lactose-free, 1 L, $4.50** — **$4.50**
Inside the rule. Added to the draft with the reason.
`reason: WITHIN_RULE · rule v3`

◇ ASK — **$5.50, or no rule saved** — **$5.50**
Request one-time approval, or skip. The rule stays the same.
`reason: ABOVE_PRICE_LIMIT · ask_above $5`

× BLOCK — **Regular milk** — **$3.29**
Conflicts with a firm rule. Omitted, and the assistant is told why.
`reason: RULE_CONFLICT · requires lactose_free`

Data changed or missing? Re-check at cart update and at checkout. Still missing → ask for a verified option or skip.

ONE ENDPOINT, THREE USERS · WHY THIS IS A PRODUCT, NOT A DEMO

Outside assistants	Instacart app	In-store shoppers
Propose swaps before checkout and relay the reason	Show the same reason to the customer at review	Apply the same rule when picking in store

03 TWO ENGINEERING LANES, FOUR WEEKS

8 engineer-weeks in total

	WEEK 1	WEEK 2	WEEK 3	WEEK 4 · SHIP
Engineer A Rules + API	Map saved rules and the 3 outcomes	Expose rule read + candidate check	Re-check price, stock, rule version	Stage behind a flag; verify hard blocks
Engineer B Cart + adapters	**Cart reaches the Instacart app** intact (the week-1 gate)	Connect draft changes; log every verdict	Wire both test adapters to the same API	Replay shared cases; fix platform drift

04 WHAT BOTH ADAPTERS MUST PASS

TEST	PASS CONDITION
Same verdict	Same candidate + same rule → same decision and reason, on both platforms
Stale quote	A changed price or rule triggers a fresh check before the cart updates
Permissions	Unlinked or wrong-account calls fail without exposing the rules
Customer review	Final cart shows every swap; the customer confirms checkout

05 FIRST PILOT SCOPE

Ship: saved item rules, read access for the linked assistant, swap checks on the draft cart.

Defer: agent-readable checkout (sprint two), agents editing rules, swaps after checkout. Instacart shoppers keep using the existing in-store replacement flow.

06 DECISION LOG

Every swap records **adapter, candidate SKU, rule version, decision, reason**, so support can trace any wrong swap in one look. This log also powers Q6.

Q4 · "QUANTITY: 2" READ TWO WAYS: WHERE IS THE PROBLEM?

The bug most likely lives in our contract: "quantity" carries no unit.

THE CALL
Treat it as a contract bug: fix once at the source, not in each agent.

WHY
Two platforms reading one field two ways means the field is underspecified. Products sell per item, per pack and by weight.

IMMEDIATE RESPONSE
Hold the ambiguous line; ask what "2" counts before the cart updates.

WE'LL WATCH
0 silent mismatches in shared tests.

01 SAME INPUT, TWO CARTS

Illustrative variants and prices

```
{"store_sku":"wheat_flour", "quantity":2}
```

PLATFORM A READS "2 UNITS"
2 × 500 g packs
1 kg
$1.80 each → **$3.60**

PLATFORM B READS "2 PACKS"
2 × 1 kg packs
2 kg
$3.20 each → **$6.40**

Same words: **2×** the flour and **+$2.80** on one line. Neither agent "misbehaved": each filled a gap we left open.

WHY ONE NUMBER CAN'T WORK · HOW GROCERIES ARE ACTUALLY SOLD

ITEM	SOLD BY	WHAT "2" COULD MEAN
Wheat flour	Pack (500 g or 1 kg)	2 packs, but of which size?
Bananas	Each or bunch	2 bananas or 2 bunches
Tomatoes	Weight (lb)	2 tomatoes or 2 lb

02 IMMEDIATE RESPONSE

Hold the ambiguous line

Ask what "2" counts and show the store's available variants before updating the cart.

If an order was already placed

Find it by trace ID, check what was bought, correct confirmed errors, refund the difference.

03 LEADING HYPOTHESIS

The MCP tool description and the API schema left the unit implicit, so each integration inferred one. **The trace confirms which layer** before we assign the cause.

We own the ambiguity
We publish the contract, so a field two agents read two ways is our bug.

Fix once, at the source
Patching one agent leaves every future agent exposed to the same gap.

No blame on partners
Blaming platforms slows the fix and makes Instacart harder to build on.

04 SIX PLACES THE UNIT CAN BE LOST

From the customer's words to the cart

MOST LIKELY · FIX HERE

CHECK
Customer's words
What was "two" meant to count?

›

CHECK
Assistant
Compare the selected SKU and raw arguments.

›

MOST LIKELY
MCP tool
Does the tool description define the unit?

›

MOST LIKELY
API schema
Does a default or mapping drop the unit?

›

CHECK
Catalog
Sold by each, pack or weight for this SKU?

›

CHECK
Cart
Stored line vs preview vs placed order.

05 FIND THE FIRST DIVERGENCE

A 3-question decision tree

IF BOTH AGENTS PRODUCED...	...THEN THE FAULT IS IN
the **same saved line**, different labels	**Display** preview or receipt wording
the **same raw request**, different validated lines	**Mapping** catalog or contract
the **same validated line**, different orders	**Checkout** cart version or re-quote
different raw requests	**Contract** the field let them differ

06 CONTROLLED REPLAY

1 Pin the inputs. Same wording, store, starting cart and catalog snapshot.
`input: "2 wheat flour" · store 214 · catalog snapshot`

2 Trace both paths. Join raw call → validated line → preview → order by trace ID.
`trace_id joins all four records`

3 Isolate the layer. Replay through each adapter, then call the API directly. Change one layer at a time.

4 Fix and prove it. Patch that layer, then re-run every shared case on both platforms until the lines match.
`exit: 0 mismatches on the shared suite`

Q4 · HOW WE FIX IT

One shared, versioned contract makes the unit explicit for every agent.

THE FIX
Require the exact SKU and sale unit. Preview the line before payment.

WHY IT HOLDS
Prevent, detect and recover in three layers, so no single check has to be perfect.

TRADE-OFF ACCEPTED
One extra prompt when an agent omits the unit.

WE'LL WATCH
Quantity corrections per **1,000** previewed carts; refunds after payment.

01 FROM AMBIGUOUS TO EXPLICIT — Illustrative

BEFORE · V1

```
{"quantity":2}
```

AFTER · V2 REQUEST

```
{"store_sku":"wheat_pack_1kg",
 "quantity":{"value":2,"unit":"pack"}}
```

CATALOG FOR THAT VARIANT

```
{"sold_by":"pack", "net_g_per_pack":1000}
```

ERROR WHEN THE UNIT IS MISSING

```
{"error":"UNIT_REQUIRED", "allowed":["1 kg pack","500 g pack"]}
```

RESULT ON BOTH PLATFORMS

```
{"line":"2 × 1 kg pack", "total_g":2000, "subtotal":6.40}
```

Same input, same line. The unit travels with the number, so the assistant cannot guess it and the cart cannot reinterpret it.

The assistant relays one plain question: **"1 kg packs or 500 g packs?"**

02 CUSTOMER CHECKOUT PREVIEW — What Maya sees

Wheat flour, 1 kg pack

2 packs — **2 kg total**

Price per pack	$3.20
Line subtotal	**$6.40**

Both agents display this **validated** line. The customer confirms in Instacart; the amount can't silently change after this.

03 FIX THE AGENT-FACING TOOL DESCRIPTION TOO — MCP tool schema

```
"quantity": { "type":"object", "required":
["value","unit"],
  "unit": { "enum":["each","pack","g"] },
  "description":"Count of the exact store variant.
   Never infer a unit; ask the customer." }
```

Agents read the description to decide what to send. Saying it plainly removes the guess at the source.

04 WHERE INSTACART ENFORCES IT

01 PREVENT
One versioned schema for API and MCP. Exact SKU, whole packs, grams for loose weight.

02 DETECT
Validate unit and value against the catalog. **Flag >3×** the usual quantity.

03 RECOVER
Bind the preview to the cart version. **Re-quote changes** before payment.

05 CONTRACT TESTS BOTH PLATFORMS MUST PASS

CASE	EXPECTED RESULT
Valid 2-pack order	Saved line and preview agree: SKU, 2 packs, 2 kg, $6.40
Missing unit / bad count	UNIT_REQUIRED; the cart stays intact
Weight on a pack SKU	Rejected; the valid pack option is returned
Quantity spike	7 packs vs a usual 2 → customer review (>3×)
Change after preview	Re-quote; the customer reviews the changed line

06 MIGRATION AND ROLLBACK

1 · Version
Require v2 for new agent writes; keep v1 only where the unit is already explicit.

2 · Replay
Compare both adapters on sample traffic with writes switched off.

3 · Enable
One adapter at a time, behind a flag.

Rollback
Any confirmed silent mismatch pauses writes for that adapter; carts fall back to customer review.

RELEASE BAR · BEFORE PAYMENT
0 silent mismatches in shared tests. Track quantity corrections per 1,000 previewed agent carts.

AFTER PAYMENT
per 1,000 quantity-related refunds per 1,000 agent orders; every confirmed mismatch traced to its cart and schema version.

Q5 · THRESHOLD OF AGENT ORDERS PER USER PER MONTH, WITH NO HISTORY

Cap autonomous orders at four per rolling 30 days.

THE CALL
4 orders / rolling 30 days, max 1 per approved weekly window, inside the spend cap.

WHY FOUR
Matches a weekly habit (**4.3** weeks a month) and caps exposure at **$320** at an $80 basket.

TRADE-OFF ACCEPTED
The 5th and any off-schedule order go back to customer checkout.

WE'LL WATCH
Unintended orders **≤0.5%**, proven on p11.

01 THE STARTING LIMIT

4 automatic orders per account, rolling 30 days

One per weekly window
Only inside the slot the customer approved, e.g. Sunday 10-12.

Inside the spend cap
All-in, with fees and tax. Over the cap goes to the customer.

Earned, not granted
Only after 3 confirmed baskets and an explicit opt-in.

Revocable in one tap
Any time, from Instacart or the assistant.

WHY ROLLING 30 DAYS, NOT A CALENDAR MONTH

Calendar cap 8 in 8 days
Rolling cap max 4
Jan 25-31, then Feb 1-7: a calendar cap resets mid-week. A rolling window counts the last 30 days, so the second four go to the customer.

WORST CASE A MONTH
$80 × 4 = $320

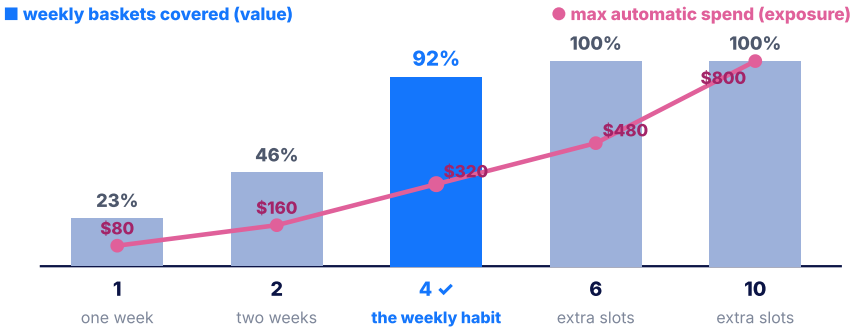
02 WHAT 30 DAYS LOOK LIKE FOR A WEEKLY CUSTOMER

Day 1	Day 8	Day 15	Day 22	Day 29
AUTO 1	AUTO 2	AUTO 3	AUTO 4	ASK · CONFIRM

A 30-day window holds about 4.3 weeks, so a 5th weekly basket can fall inside it. That one goes to customer checkout for a fresh confirmation.

03 WHY FOUR, NOT ONE OR TEN

Source: p16 · $80 cap per order



4 is the knee: it covers **92%** of a weekly customer's baskets. Going to 6 or 10 adds almost no value for a weekly customer but raises the worst case to $480-$800, before we have any evidence.

ORDERS / 30 DAYS	1	2	4	6	10
Weekly baskets covered	23%	46%	92%	100%	100%
Worst-case spend	$80	$160	$320	$480	$800
Verdict	Too little to learn	Half the habit	Start here	No gain, +50% risk	Unproven exposure

04 QUALIFY

Confirm the first three
The customer confirms 3 assistant-built baskets and can fix the list and saved rules first.

05 AUTHORISE

Opt in explicitly
Approve the list, store, address, weekly window and an all-in spend cap. Revocable any time.

06 STAY IN CONTROL

A receipt every time
Explain what was ordered and why; offer pause and a 30-minute cancel while still eligible.

07 HOW INSTACART DECIDES AT CHECKOUT

VERDICT	WHEN IT APPLIES	WHAT INSTACART DOES
✓ ACT	Valid cart and permission, unused weekly window, fewer than 4 orders in 30 days	Re-check the all-in quote, place the order, send the receipt
◇ ASK	2nd order in a window, 5th in 30 days, off-schedule, or over the cap	Route to customer checkout; approval covers this order only
× BLOCK	Permission missing or revoked, a firm rule fails, quantity still ambiguous	Stop until the issue is resolved

08 WHAT USES A SLOT

- Orders the assistant placed **without fresh confirmation**, including ones later cancelled.
- Customer-confirmed and failed attempts use **zero** slots.
- Retries share one order ID, so a retry can't spend two slots or buy twice.
- Slots are counted across all linked assistants.

Q5 · NO HISTORICAL DATA, SO WE CREATE IT

Test the cap against customer confirmation, and only grow it with evidence.

THE CALL
Run a randomized pilot: customer confirms vs assistant places up to 4.

WHY
The brief says there is no history of agent orders. A controlled test gives us the first honest baseline.

TRADE-OFF ACCEPTED
Slower expansion; we need about **600** clean orders before raising the cap.

WE'LL WATCH
Intended orders up, review effort down, harm **≤0.5%**.

01 PILOT DESIGN · SAME BASKET, DIFFERENT CHECKOUT PERMISSION

≥30 days per account

ELIGIBLE ACCOUNTS
3 confirmed baskets + explicit opt-in
Same lists, rules and weekly windows in both groups.

⇉

CONTROL · 50%
Customer confirms each checkout
The assistant prepares; the customer places every order.

TREATMENT · 50%
Assistant places up to 4
Eligible orders inside the cap; exceptions go to the customer.

Before launch: pass tests for a 5th order, retries, revoked permission and wrong units. Set the minimum improvement and feedback coverage in advance.

02 WHAT WE MEASURE

SIGNAL	MEASURE (PER ASSIGNED ACCOUNT, PER 30 DAYS)	WHAT IT TELLS US
Value	Completed, **intended** orders, including customer checkout	Does the policy add orders people wanted?
Effort	Review minutes and confirmation prompts per scheduled basket	Do customers supervise less?
Trust	Confirmed unintended orders ÷ autonomous orders; refunds and support contacts per order	Does saved time cost more wrong orders?

HOW WE VERIFY INTENT
"Did you mean for this order to be placed?"
Asked on every receipt in both groups. A "no", a cancellation in 30 minutes or a refund marked unintended counts. We audit how missing answers could hide harm.

GUARDRAILS THAT CAN STOP THE TEST ON THEIR OWN

- **Opt-outs** — Customers revoking autonomy
- **Refunds** — Per order, vs control
- **Support contacts** — Per order, vs control

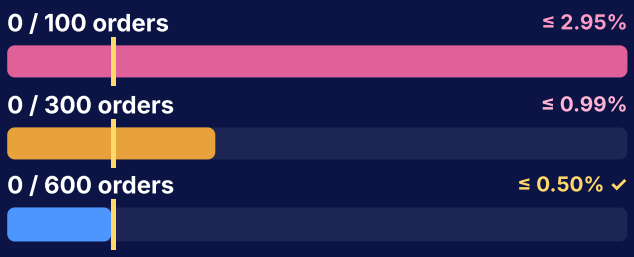
SUCCESS RULE, WRITTEN BEFORE LAUNCH

- **Intended orders up** — Treatment beats control per assigned account.
- **Effort down** — Fewer minutes and prompts per basket.
- **Trust held** — Harm bound ≤0.5%, support contacts no worse.

03 HOW MUCH EVIDENCE IS ENOUGH?

| = 0.5% ceiling · 95% confidence

With **zero** unintended orders, the true rate is at most about **3 ÷ orders** (the "rule of three").



So we need about **600** independent, intent-checked orders with no harm to show the rate is below 0.5%.

WHAT COUNTS AS HARM

- **"I didn't mean to order this"** — Said on the receipt, or a cancel within 30 minutes.
- **Outside the permission** — Wrong window, over the cap, or a firm rule broken.
- **Wrong unit at the door** — A quantity mismatch the customer confirms.

SIZING THE PILOT
At ~4 orders per account a month, 600 orders ≈ **150 account-months**: 150 treatment accounts for 1 month, or 75 for 2. One store is enough.

04 DECISION AT EACH PILOT REVIEW

Pre-agreed, so nobody argues after the data arrives

PAUSE
Stop the affected automation
A verified order outside permission pauses it. A reported unwanted order returns that customer to confirmation while we investigate.

HOLD
Keep the cap at 4
Evidence is thin, or neither value nor effort improved. Customers keep confirming.

EXPAND
Grow at four first
Value or effort improves without the other worsening, support is stable, and the harm bound is ≤0.5%. Only then test a 5th order.

Q6 · "SHOULD WE BE WORRIED THIS WILL HAPPEN TO US?"

It could. Here's how we'd know early, and what we'd do.

THE CALL
Be honest about the risk; agree evidence, owners and pause rights before launch.

WHY
False reassurance is how projects lose trust. Early, specific signals keep it.

TRADE-OFF ACCEPTED
A gated, slower rollout with visible stop points.

WE'LL WATCH
Every gate has an owner, evidence and a decision, every **4 weeks** or sooner.

01 WHAT I'D SAY TO THE CLIENT, WORD FOR WORD

"It could, and I'd take it seriously. We don't yet know why their project stalled, and our checkout handoff and quantity checks still have to pass. I'll bring you the evidence every week and flag any blocker the moment it threatens the pilot. If a gate fails, we change scope or pause that path before more customers use it. And if we stall, we say so together, early."

Honest · specific · shared ownership · no blame on the other consulting team.

WHAT I WOULD NOT SAY

× "It won't happen to us."
A promise nobody can keep.

× "Their consultants were weak."
Unverified, and it erodes trust in all of us.

× "Let's go slower to be safe."
Vague. Gates are specific.

02 AGREE TODAY: WHO CAN ACT

Instacart product lead owns scope and launch decisions.

Engagement lead (me) owns the weekly risk review and the decision record.

Engineering owner can pause an affected automation immediately, without a meeting.

03 WHY STALLS HAPPEN, AND OUR GUARD

COMMON CAUSE	OUR STRUCTURAL GUARD
Big-bang scope	One store, staples, one API
Autonomy before trust	Customer checkout first (p10-11)
No stop criteria	Pause triggers agreed upfront
Unclear owner	A named owner per gate
Hidden data gaps	Week-1 catalog audit
Public overpromise	No launch claims before p11 evidence

04 FIVE CHECKPOINTS THAT FORCE A DECISION

WHEN	OWNER	EVIDENCE TO PASS	IF IT FAILS
Week 1	Cart engineer	A prepared basket reaches customer checkout with the right items, quantities and total	Switch the sprint to checkout (p6)
Week 4	API engineer	Both adapters agree on every substitution verdict; every hard-rule violation is blocked	Fix failing cases before they appear in the demo
Week 6	Product lead	Both demo cases pass; the unit contract and preview agree across platforms	Hold the customer-checkout pilot
Before autonomy	Engineering owner	3 confirmed baskets + opt-in per account; caps, retries and revocation pass replay	Stay at customer checkout
30+ days in	Product lead	Intended orders or effort improve; support stable; harm bound **≤0.5%**	Hold the cap at 4 (p11)

05 CONTAIN — If an order goes wrong · 1 of 3

Pause the affected path
A verified out-of-permission order or a silent quantity mismatch pauses it; orders route back to customer confirmation.

06 RECOVER — 2 of 3

Help affected customers
Find the orders in the decision log (p7), correct confirmed errors, send one factual update to the sponsor.

07 RESTART — 3 of 3

Require a fresh decision
Replay the failure and the fix; the product lead approves a restart only once the checks pass.

Learn from the competitor without gossip: use verified details of their setback to stress-test our own assumptions, and escalate any confirmed overlap the same week.

LIVE PROTOTYPE · BASKET, AN AGENT-READY GROCERY CONCEPT

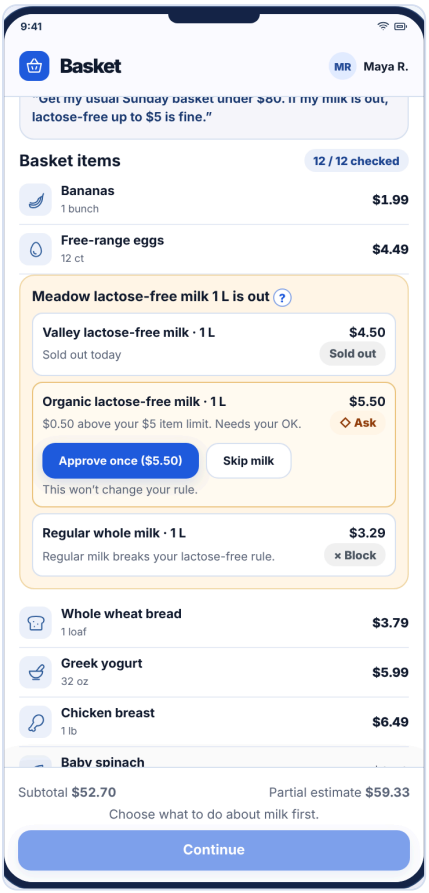
From a saved rule to an order the customer meant.

TRY IT
inmobi-prototype.vercel.app
Pick **"Show me how it works"** to see the request, rule and verdict behind every step.

60 SECONDS
▶ **Autoplay** runs the whole Sunday basket by itself.

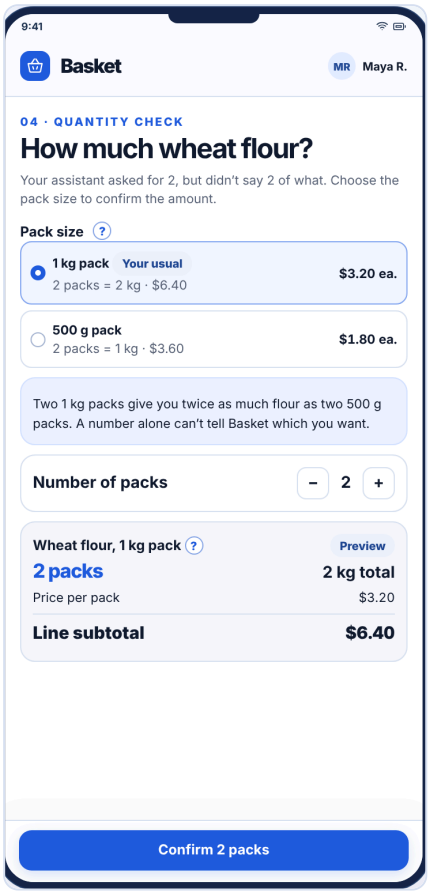
90 SECONDS
Guided tour, 5 stops, one per case question. You tap the Ask yourself.

01 Q3 ACT, ASK OR BLOCK



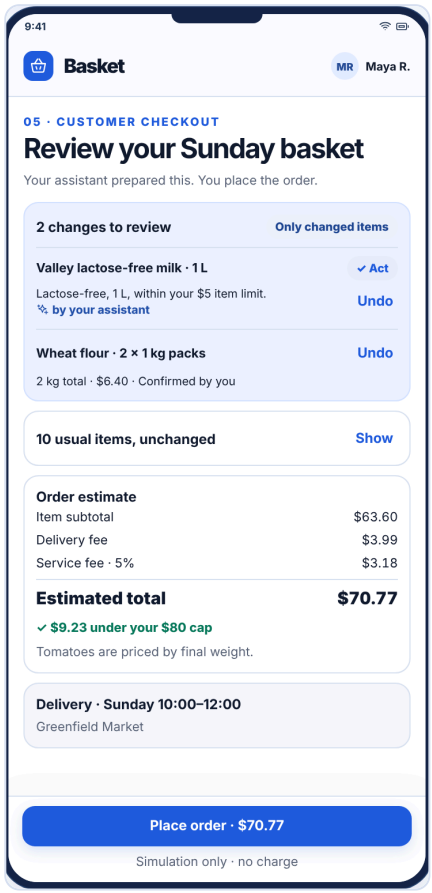
The $4.50 milk sold out, so **$5.50 waits for Maya's OK**; regular milk is blocked. Her saved $5 rule stays the same.

02 Q4 NO UNIT, NO GUESSING



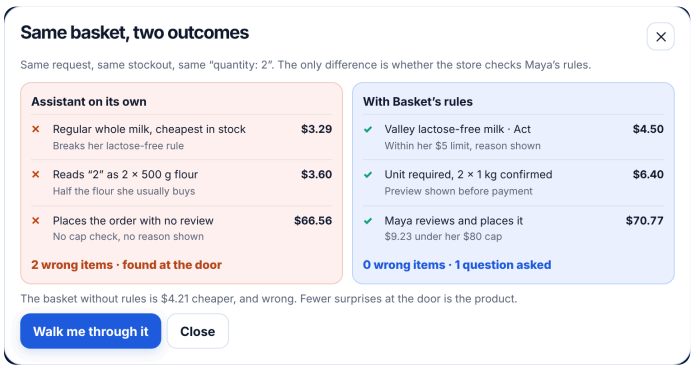
"2" is held until Maya picks the pack. The preview shows **2 × 1 kg = 2 kg, $6.40** before anything is paid.

03 Q5 THE CUSTOMER PLACES IT



Only the 2 changes need review. **$70.77, $9.23 under her $80 cap**; "I didn't mean this" feeds the Q5 harm metric.

04 THE 5-SECOND VERSION: WITHOUT RULES VS WITH BASKET

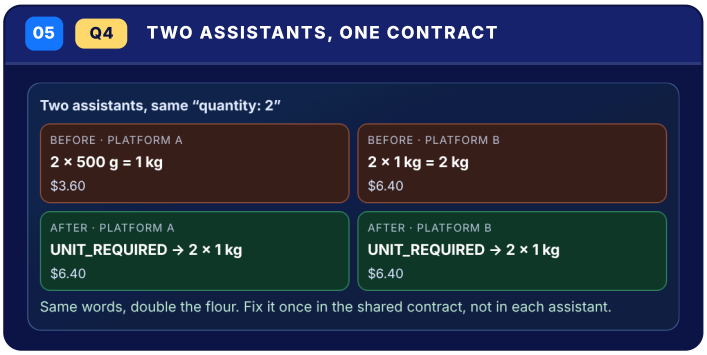


2 → 0
wrong items at the door

1
question to Maya, for the "2"

$4.21
cheaper without rules, and wrong

05 Q4 TWO ASSISTANTS, ONE CONTRACT



06 ALSO IN THE PROTOTYPE

- **What-if switches:** milk sells out · assistant asks for 7 packs (3× guard) · cap set to $65
- **Scoped permissions:** the assistant can build and swap, never pay (pilot)
- **Week-6 board demo** steps are tagged in the side panel Q2
- **Activity trace + cancel** for support and incident review Q6

CLOSE · THE 90-DAY PLAN

Ship trust first, autonomy second.

Every phase has a gate. We only move on when the evidence says the customer would have made the same call.

01 ROADMAP WITH THE GATE TO THE NEXT PHASE

1 — WEEKS 1-4 — Substitution rules

MAYA EXPERIENCES
Swaps arrive **with a reason**; regular milk never shows up.

BUILD
Saved rules, Act / Ask / Block verdict API, decision log; agent-built cart reaches the Instacart app (week-1 check).

GATE Both adapters agree; every hard-rule violation blocked

GATE OWNER · WEEK 4
API engineer

2 — WEEKS 5-6 — Unit contract + board demo

MAYA EXPERIENCES
Asked **"which pack?"** once, then it's remembered.

BUILD
v2 quantity contract (SKU + unit), checkout preview, contract tests; the week-6 board demo.

GATE Both platforms preview the same line; all demo cases pass

GATE OWNER · WEEK 6
Product lead

3 — WEEKS 7-12 — Customer-checkout pilot

MAYA EXPERIENCES
Reviews **only the changes**, then places the order herself.

LEARN
Repeat customers, one store, packaged staples. The assistant prepares; the customer places. **In parallel:** agent-readable checkout, behind the rules.

GATE Swap acceptance **≥80%**; review faster than reorder

GATE OWNER · WEEK 12
Engineering owner

4 — IF GATES PASS — Capped-autonomy test

MAYA EXPERIENCES
Her opted-in basket **arrives on schedule**, with a receipt.

TEST
Randomized 1:1 after 3 confirmed baskets and opt-in. Up to 4 orders per rolling 30 days.

GATE Value or effort up; harm bound **≤0.5%** after ~600 orders

GATE OWNER · DAY 30+
Product lead

02 DAY-90 SCORECARD · WHAT "WORKING" LOOKS LIKE

Proposed targets

≥80% — **swap acceptance**: proposed swaps kept at checkout — Q3

0 — **silent quantity mismatches** in shared contract tests — Q4

< reorder — **review time** vs a manual weekly reorder — Q1

2 — **agent platforms** live on one v2 contract — Q4

≤0.5% — **unintended orders** before any cap above 4 — Q5

03 WHAT I'D NEED FROM INSTACART

- **Catalog access**: stock, pack sizes and ingredient data for one pilot store.
- **A pilot store** with a working customer-checkout path.
- **A named product owner** who can pause, approve or reject a rollout.
- **Two agent platforms** willing to integrate against the v2 contract.

04 WHAT I'D VALIDATE FIRST

- Does the checkout handoff preserve exact lines and the all-in price?
- How often are pack and dietary data missing in the catalog?
- Will customers opt in, and reliably report intent on the receipt?
- What does a "good" swap look like by category (dairy vs produce)?

05 THE SIX CALLS, IN ONE BREATH

Q1	Q2	Q3	Q4	Q5	Q6
Weekly baskets first	Keep the date, change the demo	Substitution rules before checkout	Make the unit explicit	Cap at 4 / 30 days	Gates with owners

THANK YOU

See the whole argument run in 60 seconds →

inmobi-prototype.vercel.app · Varun Kale · ED22B018 · IIT Madras

Appendix

Sources and notes on every figure →

APPENDIX · SOURCES

Every external number, one click from its source.

Tap any title to open the source in your browser. Tap a page number to jump to where it is used.

01 EXTERNAL SOURCES

All links accessed November 2026

#	SOURCE	WHAT IT SUPPORTS	USED ON
1	**Which?, "The silliest supermarket substitutions revealed"** 12 Mar 2026 · Nov 2025 survey of 3,552 UK adults, 1,417 online shops	1 in 5 online grocery shoppers got a substitution in their latest shop	p3 · p4
2	**Knight & Mitrofanov, "Disclosing Low Product Availability"** *Management Science* (2025) · Instacart field experiment	840,000+ customers; -2.9% replacements, -7.5% poor ratings, -3.7% refunds	p4 · p6
3	**Instacart Help Center, "Shop with third-party AI apps"** How outside assistants hand a cart to Instacart checkout	Real-world check: customers can already pay for an agent-built cart, so Q3's human-pay path is realistic	p6
4	**Instacart Docs, "Replacement instructions"** Customers save replacement choices that carry over to future orders	Saved preferences exist but agents cannot read or apply them yet	p6
5	**Instacart Docs, "Replacements flow"** Instacart Connect post-checkout guide	Where an Act / Ask / Block verdict plugs into today's flow	p7
6	**Instacart Docs, "Product quantity"** Items sold by each, by package or by weight	Why "quantity: 2" needs a unit (each, pack, kg)	p8 · p9
7	**Model Context Protocol, "Tools" specification** Version 2025-11-25 · tool names, descriptions, input schemas	The tool description is part of the contract agents read	p9
8	**NIST/SEMATECH Dataplot, "Exact binomial"** One-sided exact upper confidence limit for a proportion	0 / 600 orders → true rate ≤ 0.5% at 95%	p11
9	**Live prototype, inmobi-prototype.vercel.app** Built for this case · not affiliated with Instacart	Every flow in the deck, clickable	p13

02 HOW THE DECK'S NUMBERS ARE WORKED OUT

FIGURE	WORKING	PAGE
~52 occasions	One basket a week × 52 weeks	p4
$3.60 vs $6.40	2 × 500 g at $1.80 vs 2 × 1 kg at $3.20	p8
92% coverage	min(4, 4.35) ÷ 4.35 weekly baskets in 30 days	p10
$320 worst case	$80 cap × 4 orders per rolling 30 days	p10
≤ 0.5% at 600	$1 - 0.05^{1/n} \approx 3 \div n$, zero events in n orders	p11
≤ 2.95% · ≤ 0.99%	Same bound at 100 and 300 clean orders	p11
2 × 4	Two engineers × four weeks, from the brief	p7
$70.77, $9.23 under	Prototype basket total vs the $80 cap	p13
150 account-months	600 orders ÷ ~4 orders per account a month	p11

03 WHAT IS ILLUSTRATIVE, AND WHAT COMES FROM THE BRIEF

- **From the case brief:** the 12-month agent horizon, the six-week board date, the three-month concern, two engineers for four weeks, and no history of agent-placed orders.
- **Illustrative:** Maya's basket, product names, prices, the $80 cap, API fields and every dashboard figure in the prototype.
- **Proposed pilot settings, not results:** 80% swap acceptance, 70% demo reuse, the 0.5% harm ceiling, the 3× quantity flag and the four-order limit.
- **My judgment:** the arena map on p4 is directional. Checkpoint owners on p12 are roles, not named people.
- **Directional evidence:** the Which? data is from the UK; the US pilot measures its own baseline before any target is fixed.

04 SEE IT LIVE

inmobi-prototype.vercel.app

Every flow in this deck, clickable. Built for this case; not affiliated with Instacart.

← Back to the agenda